

The Economist explains

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How countries calculate their GDP



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WHEN unveiling Britain's annual budget (<http://www.economist.com/node/21599379>) on March 19th George Osborne, the chancellor of the exchequer, crowed that the British economy was forecast to grow at an annualised rate of 2.7% in the first quarter of 2014, the fastest in the rich world. His critics countered that whereas output in America and Germany has already topped the pre-crisis peak, Britain's will not get there until later this year. The data-point at issue in both cases is gross domestic product, or GDP, the total value of all goods and services produced within an economy each year. GDP is of critical economic importance; thousands of economists use estimates of the total amount spent or (equivalently) earned each year in their research. Governments also rely heavily on the figure, to shape policy or determine how much public

spending is affordable. Yet GDP seems an impossibly complex thing to measure in a modern economy. How do countries calculate it?

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British and French economists began to estimate the total income earned in their economies in the late 16th and early 17th centuries, primarily to help their sovereigns find better ways to raise tax revenue. But proper estimates were not regularly produced until the early 20th century, when modern statistical techniques and the demands of total war encouraged governments to take a greater interest in national accounting. In most rich economies annual estimates are available from the 1930s, thanks to the travails of the Depression. Simon Kuznets, a Russian emigrant to America, is credited with creating the first true GDP estimate, for delivery to America's Congress in 1934. Governments of the day were determined to manage economic ups and downs and required regularly updated figures to do so. The outbreak of the second world war, and its consequent economic demands, pushed the task of economic measurement firmly into government hands. From then on, GDP estimates were produced by government statistical offices.

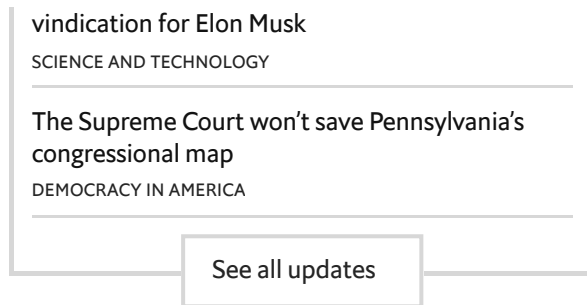
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Output can be measured in three (theoretically equivalent) ways: by adding up all the money spent each year, by adding up all the money earned each year, or by



adding up all the value added each year. Some economies, including Britain, combine all three methods into a single GDP figure, whereas others, like America, produce different statistics for each.

(American GDP is estimated via the

spending approach; GDI, or gross domestic income, by the income approach.) Data are gathered from many small surveys. America's Bureau of Economic Analysis draws data (<http://www.economist.com/news/finance-and-economics/21582498-america-has-changed-way-it-measures-gdp-boundary-problems>) from surveys of manufacturers, builders and retailers, as well as from trade and financial flows, among other sources. These data are used to estimate the components of GDP, such as total investment and net exports. Owing to the demand for timely data, preliminary estimates are released and subsequently revised as more information is obtained. At longer intervals GDP statistics are given bigger overhauls, both to revise data and to recalibrate the underlying statistical models.

For all its uses, GDP is an imperfect measure. Different flavours of the statistic are more or less useful for different purposes. Real, or inflation-adjusted, GDP is needed to compare figures across time periods, while GDP per person (<http://www.economist.com/node/10852462>) is best for understanding how individual incomes are evolving. Some reckon (<http://www.economist.com/debate/days/view/501>) that GDP can mislead. Money spent on activities that generate pollution, or on medical treatments that don't work, adds to GDP but does not reflect any improvement in national welfare. Indeed, a few outside-the-box thinkers reckon it ought to be scrapped entirely. In 1972 the king of Bhutan announced a plan to focus on "gross national happiness". In recent years a few rich-world leaders have pushed efforts (<http://www.economist.com/node/21548213>) to study whether a happiness statistic could prove useful. In the meantime, money spent on such projects will be counted towards good old GDP.

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